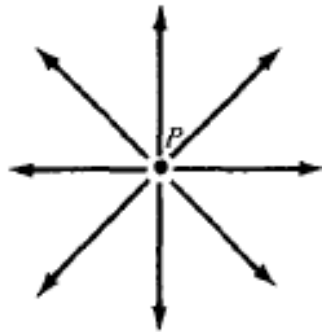


CURL AND DIVERGENCE

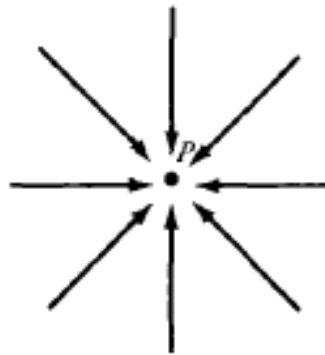
DIVERGENCE OF A VECTOR AND DIVERGENCE THEOREM

- The divergence of $\mathbf{A}$ at a given point P is *the outward flux per unit volume as the volume shrinks about P .*

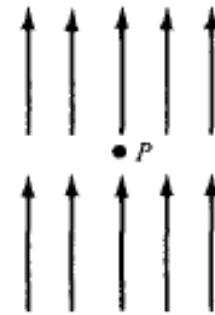
$$\text{div } \mathbf{A} = \nabla \cdot \mathbf{A} = \lim_{\Delta V \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta V}$$



Positive divergence



Negative divergence



Zero divergence

- where Δv is the volume enclosed by the closed surface S in which P is located.
- *Physically*, we may regard the divergence of the vector field $\mathbf{A}$ at a given point as a measure of how much the field diverges or emanates from that point.

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Divergence theorem

- The **divergence theorem** states that the **total outward flux of a vector field $\mathbf{A}$ through the closed surface S is the same as the volume integral of the divergence of $\mathbf{A}$.**

$$\oint_S \mathbf{A} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{A} \, dv$$

- The theorem applies to any volume V bounded by the closed surface S

Example

Determine the divergence of these vector fields:

(a) $\mathbf{P} = x^2 y z \mathbf{a}_x + x z \mathbf{a}_z$

(b) $\mathbf{Q} = \rho \sin \phi \mathbf{a}_\rho + \rho^2 z \mathbf{a}_\phi + z \cos \phi \mathbf{a}_z$

(c) $\mathbf{T} = \frac{1}{r^2} \cos \theta \mathbf{a}_r + r \sin \theta \cos \phi \mathbf{a}_\theta + \cos \theta \mathbf{a}_\phi$

Solution

$$\begin{aligned} \text{(a)} \quad \nabla \cdot \mathbf{P} &= \frac{\partial}{\partial x} P_x + \frac{\partial}{\partial y} P_y + \frac{\partial}{\partial z} P_z \\ &= \frac{\partial}{\partial x} (x^2 y z) + \frac{\partial}{\partial y} (0) + \frac{\partial}{\partial z} (x z) \\ &= 2xyz + x \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \nabla \cdot \mathbf{Q} &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho Q_\rho) + \frac{1}{\rho} \frac{\partial}{\partial \phi} Q_\phi + \frac{\partial}{\partial z} Q_z \\ &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho^2 \sin \phi) + \frac{1}{\rho} \frac{\partial}{\partial \phi} (\rho^2 z) + \frac{\partial}{\partial z} (z \cos \phi) \\ &= 2 \sin \phi + \cos \phi \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad \nabla \cdot \mathbf{T} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 T_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (T_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (T_\phi) \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} (\cos \theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (r \sin^2 \theta \cos \phi) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (\cos \theta) \\ &= 0 + \frac{1}{r \sin \theta} 2r \sin \theta \cos \theta \cos \phi + 0 \\ &= 2 \cos \theta \cos \phi \end{aligned}$$

CURL OF A VECTOR AND STOKES'S THEOREM

- we defined the circulation of a vector field $\mathbf{A}$ around a closed path L as the integral.

$$\oint_L \mathbf{A} \cdot d\mathbf{l}.$$

- The **curl of $\mathbf{A}$** is an axial (or rotational) vector whose magnitude is **the maximum circulation** of $\mathbf{A}$ per unit area as the area tends to zero and whose direction is the normal direction of the area when the area is oriented so as to make the circulation maximum."

$$\text{curl } \mathbf{A} = \nabla \times \mathbf{A} = \left(\lim_{\Delta S \rightarrow 0} \frac{\oint_L \mathbf{A} \cdot d\mathbf{l}}{\Delta S} \right)_{\text{max}} \mathbf{a}_n$$

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{a}_\rho & \rho \mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{a}_r & r \mathbf{a}_\theta & r \sin \theta \mathbf{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}$$

Stokes' theorem

- **Stokes's theorem states that the circulation of a vector field $\mathbf{A}$ around a (closed) path L is equal to the surface integral of the curl of $\mathbf{A}$ over the open surface S *bounded by* L provided that $\mathbf{A}$ and curl of $\mathbf{A}$ are continuous on S**

$$\oint_L \mathbf{A} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

CLASSIFICATION OF VECTOR FIELDS

- A vector field $\mathbf{A}$ is said to be solenoidal (or divergenceless) if $\nabla \cdot \mathbf{A} = 0$.

$$\oint_S \mathbf{A} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{A} \, dv = 0$$

- vector field $\mathbf{A}$ is said to be irrotational (or potential) if $\nabla \times \mathbf{A} = 0$.

$$\int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint_L \mathbf{A} \cdot d\mathbf{l} = 0$$

- Thus in an irrotational field A , the circulation of A around a closed path is identically zero.
- This implies that the line integral of A is independent of the chosen path.
- Therefore, an irrotational field is also known as a ***conservative field***.
- *Examples of irrotational fields include the electrostatic field and the gravitational field.*

END